Sri Lanka Institute of Information Technology

IT3021 - Data Warehousing & Business Intelligence

Assignment 2

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1. Introduction

In this project, we will design and implement a Data Warehouse and Business Intelligence for analyzing loan-related data. This data is then aggregated and processed, from where the system finds a meaningful value to it and helps in recommendations.

The solution has a logical method which consists

- Star schema for data warehouse design
- Implementation of ETL processes
- OLAP cube Development using SSAS
- Visualization using Power BI

This project corresponds to a dataset that contains sales information such as customer, sales person, product, delivery location and profit from the sales. By structuring this information based on a dimensional model, it allows for the easy querying and analytical processing capabilities of the system.

The project shows how to leverage modern data warehousing techniques to,

- Improve data organization
- Enable fast analytical queries
- Support business

1.1 Project Objectives

This project aims at achieving the following goals,

- A star schema for designing a data warehouse
- For OLAP, you use dimension and fact tables
- So now to create a SSAS cube Mulitdimensional Analysis
- To enable better insights and measures hierarchies
- You will learn to create interactive Power BI dashboards
- Augmented data visualization to make informed decisions

1.2 Scope of the Project

This project looks at sales data from different angles, like sellar, place, product type and customers. Users can look at important performance indicators like the amount of money funded, the amount of the sales and the time it takes to process a transaction.

The scope includes:
• Design and build a data warehouse
• Making a cube with SSAS
• Making reports with Power BI

2. Data Source

For this project, we used the data warehouse we built earlier as our main source of information. This warehouse stored organized data about sales, making it easier to analyze. To work with this data, we relied on Microsoft Excel and Power BI. These tools allowed us to perform various analytical tasks using the dataset provided. By combining the structured data from the warehouse with the capabilities of Excel and Power BI, we were able to gain insights into the loan transactions and execute our analytical operations effectively.

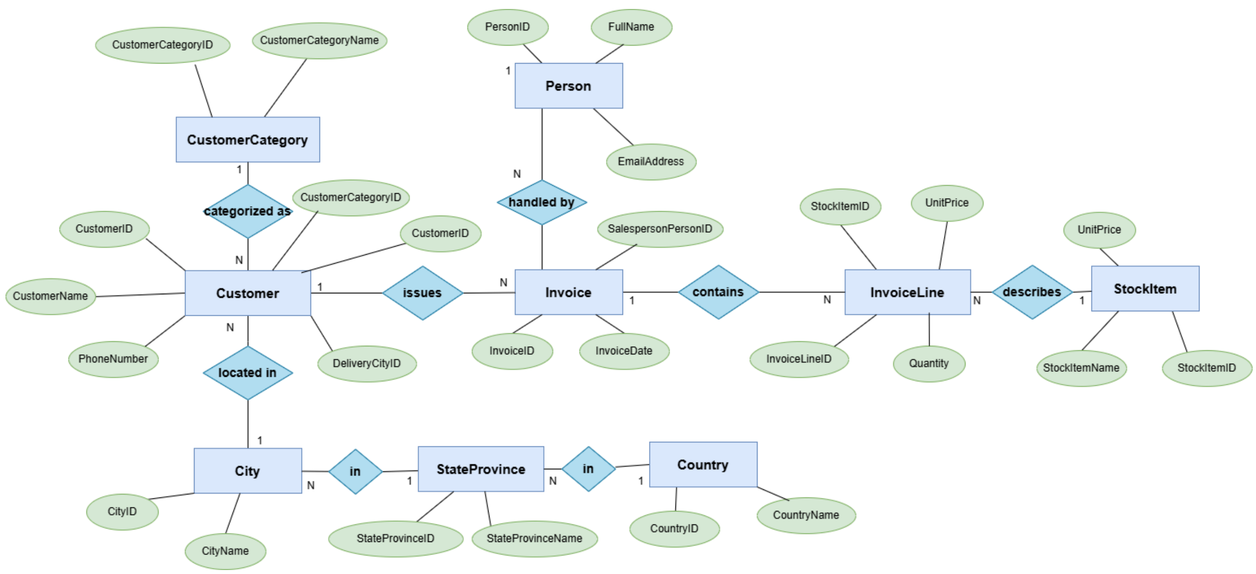
The information we got was about the sales and details of the sales. It was given to us in a special format, with two kinds of tables as fact tables and dimension tables. We used this to make a data warehouse and we set it up in a way that's called a star schema model. This makes it easy to look at the data from different angles and get the information we need.

The fact table gave us the numbers we needed to understand the dataset.

The dimension tables also gave us the details we needed to analyze things. They had all the descriptions and other information that was necessary for our purpose.

3. Data Warehouse Design

3.1 ER diagram



3.2 Dimensional Model

The data warehouse follows a star schema architecture, where a central fact table is connected to multiple dimension tables.

- The Fact_Sales table contains measurable data such as Sales Amount, Quantity, Profit, and Cost.
- The dimension tables provide descriptive information:
    - DimCustomer → customer details
    - DimProduct → product information
    - DimSalesperson → employee details
    - DimDate → time-related attributes

This structure improves query performance and simplifies analytical processing.

3.2.1 DimCustomer

```
CREATE TABLE dbo.DimCustomer (
    CustomerSK int IDENTITY(1,1) NOT NULL,
    AlternateCustomerID int NULL,
    CustomerName nvarchar(50) NULL,
    CustomerCategory nvarchar(50) NULL,
    CustomerCity nvarchar(50) NULL,
    CustomerStateProvince nvarchar(50) NULL,
    CustomerCountry nvarchar(60) NULL,
    CustomerRegion nvarchar(30) NULL,
    CustomerContinent nvarchar(30) NULL,
    CustomerPhoneNumber nvarchar(50) NULL,
    CustomerFaxNumber nvarchar(50) NULL,
    StartDate datetime NULL,
    EndDate datetime NULL,
    InsertDate datetime NULL,
    ModifiedDate datetime NULL,
    CONSTRAINT PK_DimCustomer PRIMARY KEY CLUSTERED (CustomerSK)
);
```

Results | Messages

	CustomerSK	AlternateCustomerID	CustomerName	CustomerCategory	CustomerCity	CustomerStateProvince	CustomerCountry	CustomerRegion	CustomerContine
262	262	13	Tailspin Toys (Stonefort	Agent	Abanda	Alabama	United States	Americas	North America
263	263	12	Tailspin Toys (Biscay	Agent	Abanda	Alabama	United States	Americas	North America
264	264	11	Tailspin Toys (Devault	Agent	Abanda	Alabama	United States	Americas	North America
265	265	2	Tailspin Toys (Sylvanite	Agent	Abanda	Alabama	United States	Americas	North America
266	266	3	Tailspin Toys (Peeples Val...	Agent	Abanda	Alabama	United States	Americas	North America
267	267	4	Tailspin Toys (Medicine L...	Agent	Abanda	Alabama	United States	Americas	North America
268	268	5	Tailspin Toys (Gasport	Agent	Abanda	Alabama	United States	Americas	North America
269	269	6	Tailspin Toys (Jessie	Agent	Abanda	Alabama	United States	Americas	North America
270	270	7	Tailspin Toys (Frankewing	Agent	Abanda	Alabama	United States	Americas	North America
271	271	8	Tailspin Toys (Bow Mar	Agent	Abanda	Alabama	United States	Americas	North America

3.2.2 DimProduct

```
CREATE TABLE dbo.DimProduct (
  ProductSK int IDENTITY(1,1) NOT NULL,
  AlternateStockItemID int NULL,
  StockItemName nvarchar(100) NULL,
  SupplierID int NULL,
  ColorID int NULL,
  Brand nvarchar(50) NULL,
  Size nvarchar(20) NULL,
  UnitPrice decimal(18,2) NULL,
  TaxRate decimal(18,3) NULL,
  RecommendedRetailPrice decimal(18,2) NULL,
  TypicalWeightPerUnit decimal(18,3) NULL,
  IsChillerStock bit NULL,
  CONSTRAINT PK_DimProduct PRIMARY KEY CLUSTERED (ProductSK)
);
```

Results Messages

	ProductSK	AlternateStockItemID	StockItemName	SupplierID	ColorID	Brand	Size	UnitPrice	TaxRate	RecommendedRetailPrice	TypicalW
514	514	60	RC toy sedan car with remote control (Blue) 1/5...	10	4	Nort...	1/5...	25.00	15.000	37.38	1.500
515	515	61	RC toy sedan car with remote control (Green) 1/...	10	NULL	Nort...	1/5...	25.00	15.000	37.38	1.500
516	516	62	RC toy sedan car with remote control (Yellow) 1...	10	36	Nort...	1/5...	25.00	15.000	37.38	1.500
517	517	63	RC toy sedan car with remote control (Pink) 1/5...	10	NULL	Nort...	1/5...	25.00	15.000	37.38	1.500
518	518	64	RC vintage American toy coupe with remote co...	10	28	Nort...	1/5...	30.00	15.000	44.85	1.700
519	519	65	RC vintage American toy coupe with remote co...	10	3	Nort...	1/5...	30.00	15.000	44.85	1.700
520	520	66	RC big wheel monster truck with remote control ...	10	3	Nort...	1/5...	45.00	15.000	67.28	1.800
521	521	67	Ride on toy sedan car (Black) 1/12 scale	10	3	Nort...	1/1...	230.00	15.000	343.85	15.000
522	522	68	Ride on toy sedan car (Red) 1/12 scale	10	28	Nort...	1/1...	230.00	15.000	343.85	15.000
523	523	69	Ride on toy sedan car (Blue) 1/12 scale	10	4	Nort...	1/1...	230.00	15.000	343.85	15.000

3.2.3 DimSalesPerson

```
CREATE TABLE dbo.DimSalesperson (

  SalespersonSK int IDENTITY(1,1) NOT NULL,

  AlternatePersonID int NOT NULL,

  FullName nvarchar(100) NOT NULL,

  PreferredName nvarchar(100) NULL,

  EmailAddress nvarchar(256) NULL,

  PhoneNumber nvarchar(50) NULL,

  IsEmployee bit NULL,

  IsSalesperson bit NULL,

  CONSTRAINT PK_DimSalesperson PRIMARY KEY CLUSTERED (SalespersonSK)

);
```

Results | Messages

	SalespersonSK	AlternatePersonID	FullName	PreferredName	EmailAddress	PhoneNumber	IsEmployee	IsSalesperson
1	5557	1	Data Conversion Only	Data Conversion Only	NULL	NULL	0	0
2	5558	2	Kayla Woodcock	Kayla	kaylaw@wideworldimporters.com	(415) 555-0102	1	1
3	5559	3	Hudson Onslow	Hudson	hudsono@wideworldimporters.com	(415) 555-0102	1	1
4	5560	4	Isabella Rupp	Isabella	isabellar@wideworldimporters.com	(415) 555-0102	1	0
5	5561	5	Eva Muirden	Eva	evam@wideworldimporters.com	(415) 555-0102	1	0
6	5562	6	Sophia Hinton	Sophia	sophiah@wideworldimporters.com	(415) 555-0102	1	1
7	5563	7	Amy Trefl	Amy	amyt@wideworldimporters.com	(415) 555-0102	1	1
8	5564	8	Anthony Grosse	Anthony	anthonyg@wideworldimporters.com	(415) 555-0102	1	1
9	5565	9	Alica Fatnowna	Alica	alicaf@wideworldimporters.com	(415) 555-0102	1	0
10	5566	10	Stella Rosenhain	Stella	stellar@wideworldimporters.com	(415) 555-0102	1	0

3.2.4 DimDate

Results | Messages

	DayOfMonth	DaySuffix	DayName	DayOfWeekUSA	DayOfWeekUK	DayOfWeekInMonth	DayOfWeekInYear	DayOfQuarter	DayOfYear	WeekOfMonth	WeekOfQuarter
1	1	1st	Monday	2	1	1	1	1	1	1	1
2	2	2nd	Tuesday	3	2	1	1	1	2	1	1
3	3	3rd	Wednesday	4	3	1	1	1	3	1	1
4	4	4th	Thursday	5	4	1	1	1	4	1	1
5	5	5th	Friday	6	5	1	1	1	5	1	1
6	6	6th	Saturday	7	6	1	1	1	6	1	1
7	7	7th	Sunday	1	7	1	1	1	7	2	1
8	8	8th	Monday	2	1	2	2	2	8	2	2
9	9	9th	Tuesday	3	2	2	2	2	9	2	2
10	10	10th	Wednesday	4	3	2	2	2	10	2	2

3.2.5 Fact_Sales

The Fact_sales is the central table in the data warehouse that stores quantitive data related to sales transactions. It contains measurable values such as Sales Amount, Quantity, Cost Amount, Profit Amount, Tax Amount and Unit Price.

Through foreign keys, this table is linked to other dimension tables, including Customer, Product, Salesperson and Date.

These relationships allow the data to be analyzed from different perspectives, enabling multidimensional analysis in the OLAP cube.

This dimension enables time-based analysis and supports hierarchical navigation

Year → Quarter → Month → Day

It plays a key role in identifying trends, patterns and seasonal variations in sales data

Results | Messages

	'rice	TaxAmount	SalesAmount	CostAmount	ProfitAmount	InsertDate	ModifiedDate	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	0	4.80	36.80	16.80	20.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-29 11:38:00.000	24.53
2	0	259.20	1987.20	1699.20	288.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-30 14:55:00.000	51.82
3	00	299.25	2294.25	1384.25	910.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-31 08:12:00.000	69.10
4	0	3.75	28.75	22.75	6.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-04-01 11:29:00.000	96.38
5	0	9.75	74.75	32.25	42.50	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-26 14:46:00.000	-44.33
6	0	28.80	220.80	100.80	120.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-27 08:03:00.000	-27.05
7	0	19.20	147.20	51.20	96.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:20:00.000	0.23
8	0	11.70	89.70	38.70	51.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-29 14:37:00.000	27.52
9		79.92	612.72	267.12	345.60	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-30 08:54:00.000	45.80
10	0	26.25	201.25	113.75	87.50	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-31 11:11:00.000	72.08

Results | Messages

	InvoiceID	InvoiceLineID	InvoiceDateKey	ConfirmedDeliveryDateKey	CustomerKey	ProductKey	SalespersonKey	Quantity	UnitPrice	TaxAmount	SalesAmount	CostAmount	ProfitAmoun
1	29876	97112	20140718	20140719	63	9	5559	1	32.00	4.80	36.80	16.80	20.00
2	29877	97116	20140718	20140719	184	149	5571	96	18.00	259.20	1987.20	1699.20	288.00
3	29877	97117	20140718	20140719	184	73	5571	7	285.00	299.25	2294.25	1384.25	910.00
4	29878	97118	20140718	20140719	572	110	5563	1	25.00	3.75	28.75	22.75	6.00
5	29878	97119	20140718	20140719	572	33	5563	5	13.00	9.75	74.75	32.25	42.50
6	29878	97120	20140718	20140719	572	5	5563	6	32.00	28.80	220.80	100.80	120.00
7	29879	97122	20140718	20140719	590	126	5558	4	32.00	19.20	147.20	51.20	96.00
8	29879	97121	20140718	20140719	590	54	5558	6	13.00	11.70	89.70	38.70	51.00
9	29880	97124	20140718	20140719	636	195	5570	144	3.70	79.92	612.72	267.12	345.60
10	29880	97123	20140718	20140719	636	62	5570	7	25.00	26.25	201.25	113.75	87.50

4. SSAS Cube Implementation

After successfully designing and loading the data warehouse using SSIS, SQL Server Analysis Services (SSAS) was used to create an OLAP (Online Analytical Processing) cube. This cube allows for multidimensional analysis and supports efficient querying for business intelligence and reporting.

4.1 Creating the SSAS Project

A new SSAS Multidimensional project was created in Visual Studio. The data source was set to the data warehouse database.

4.2 Building the Data Source View (DSV)

The Data Source View was created using the following tables from the data warehouse:

• FactSales

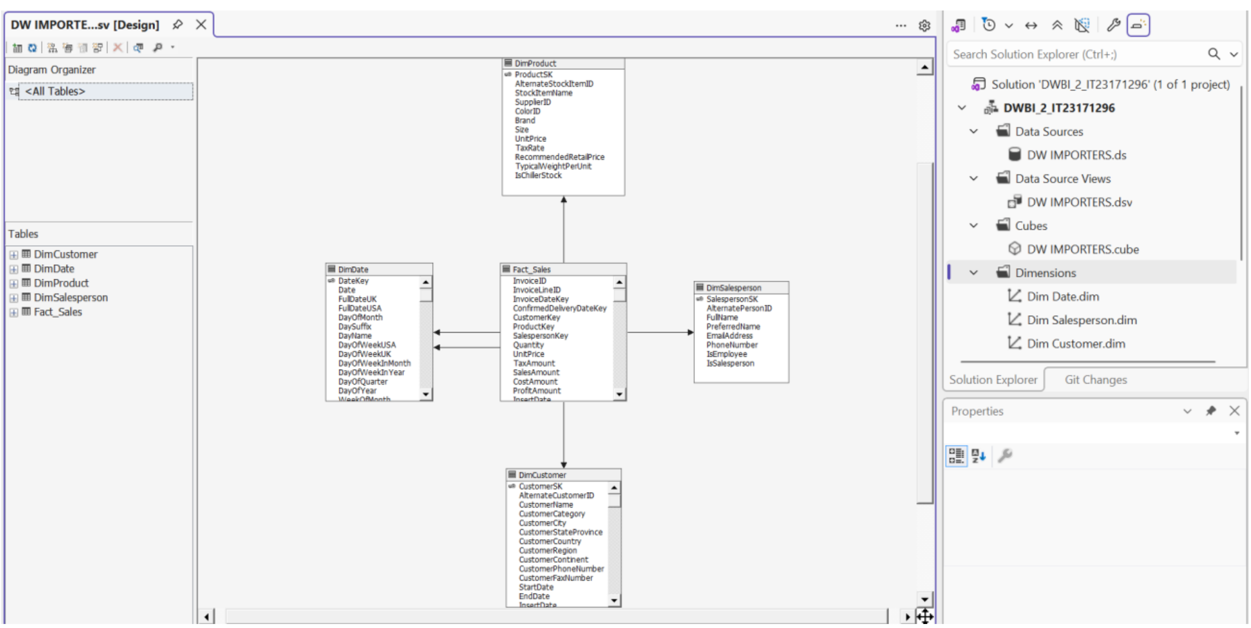
• DimCustomer

• DimSalesperson

• DimProduct

• DimDate

The relationships between these fact tables and dimension tables were verified in the DSV



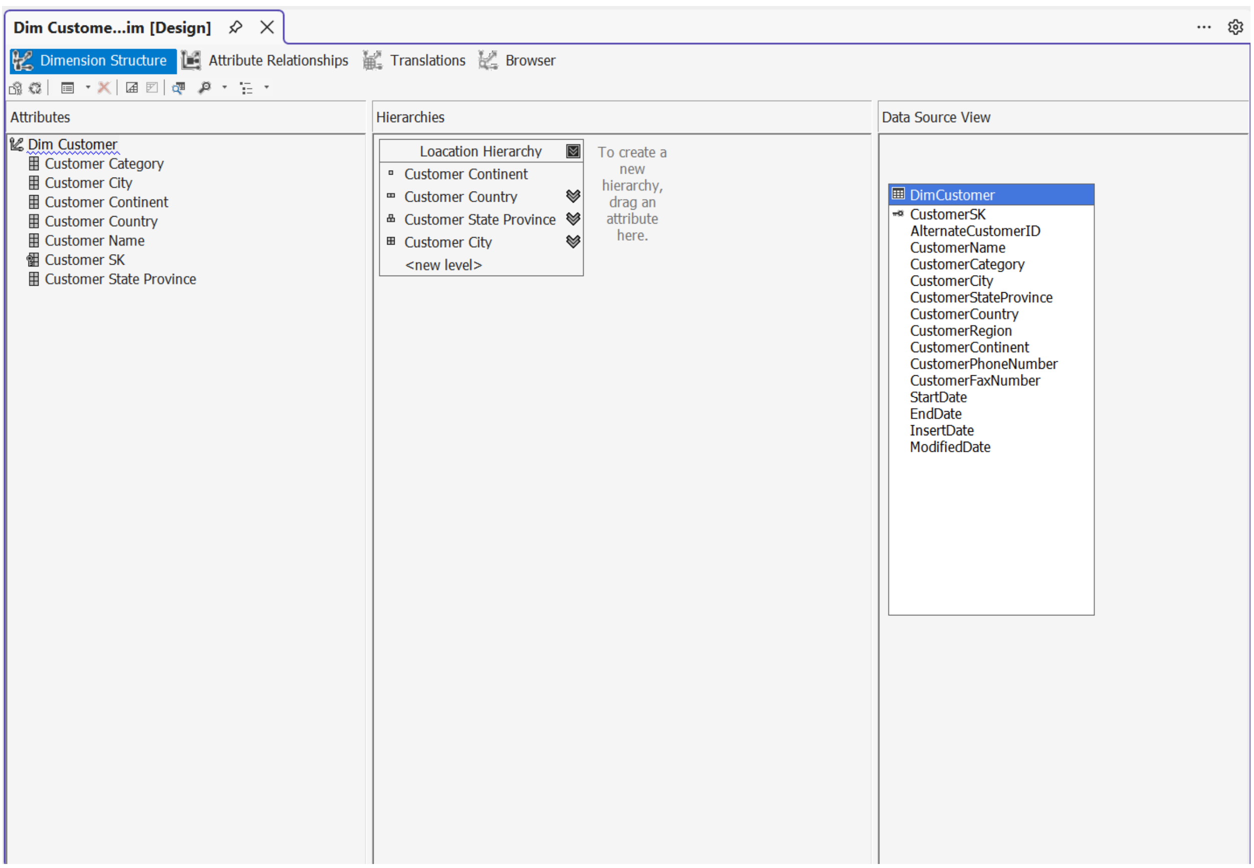
4.3 Creating Dimensions

From each dimension table, dimensions were made. We carefully chose the relevant key columns and name columns to make sure that client tools could browse easily.

4.3.1 DimCustomer

The DimCustomer dimension was designed to analyze customer-related data. Attributes such as Customer Name, Customer Country, Customer State Province, Customer City, and Customer Category were included.

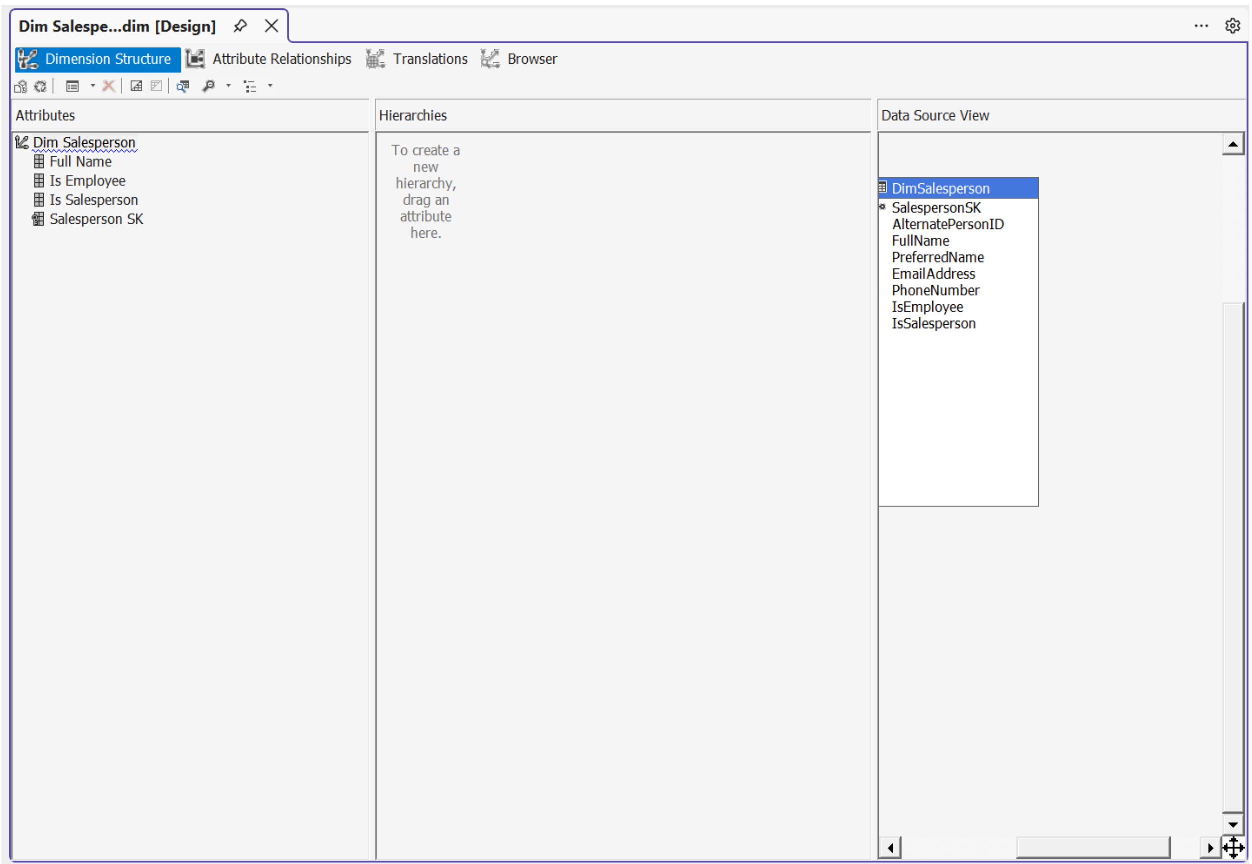
A Location Hierarchy (Country → State → City) was created to support geographical analysis.



4.3.2 DimSalesPerson

The DimSalesperson dimension contains information about sales employees. Attributes such as Full Name, Is Salesperson, and Is Employee were included.

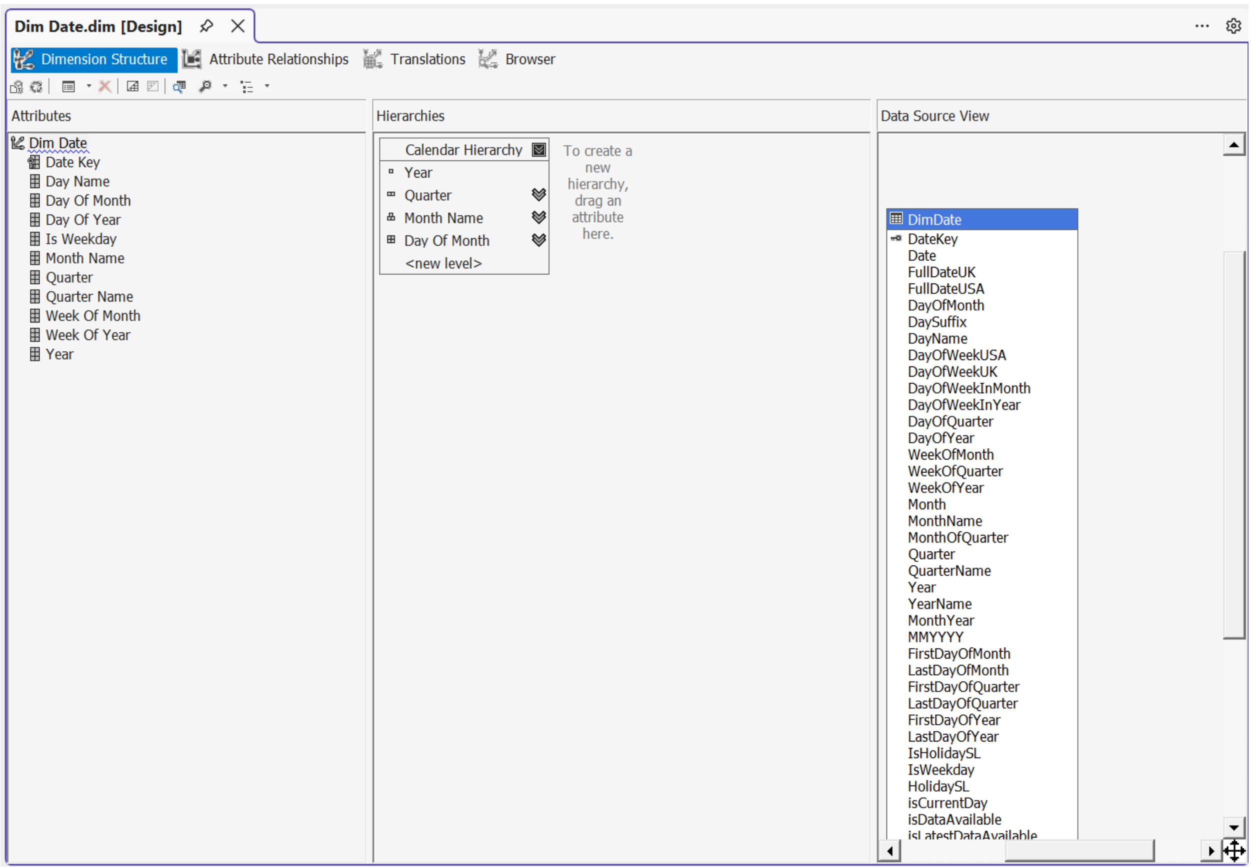
This dimension allows analysis of sales performance based on individual salespersons.



4.3.3 DimDate

The DimDate dimension was created to enable time-based analysis. Attributes such as Year, Quarter, Month Name, Day of Month, and Day Name were included.

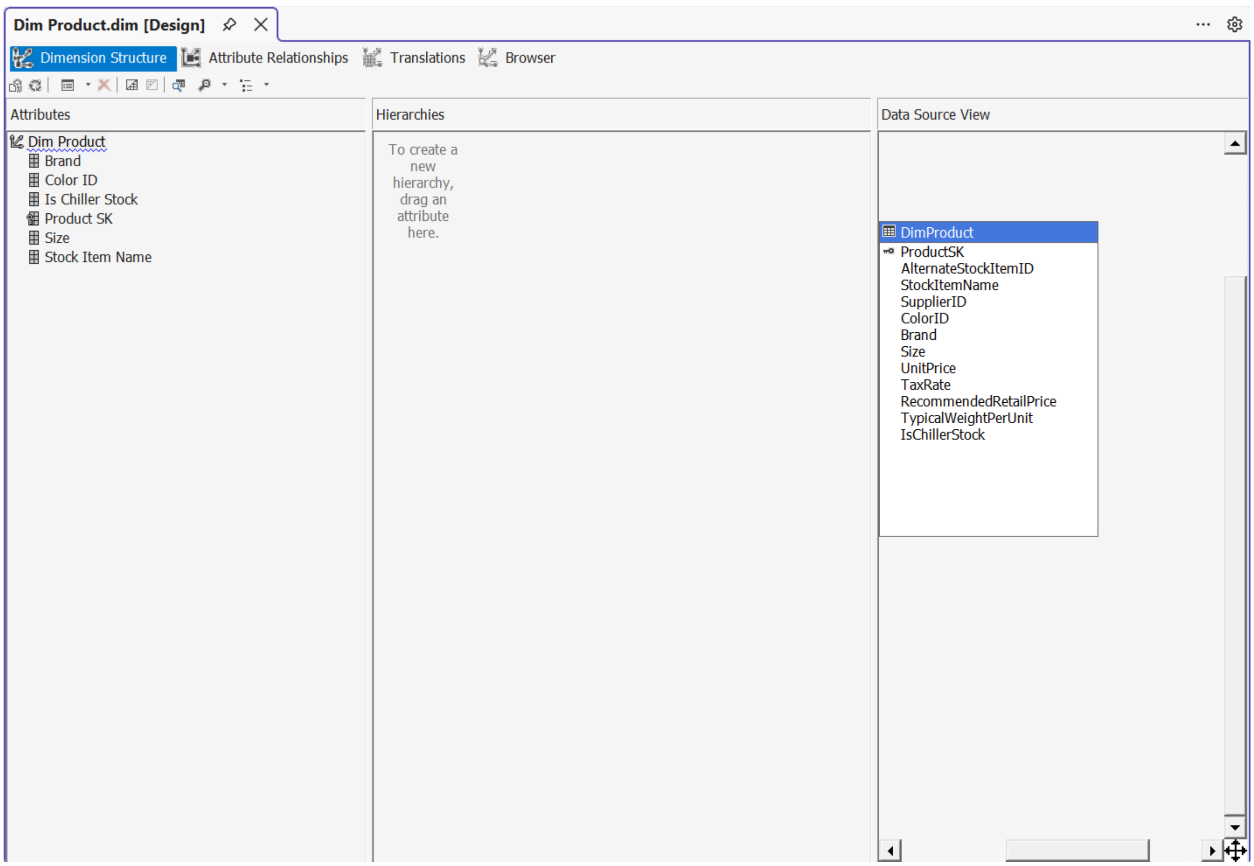
A Calendar Hierarchy (Year → Quarter → Month → Day) was created to support drill-down operations in the cube.



4.3.4 DimProduct

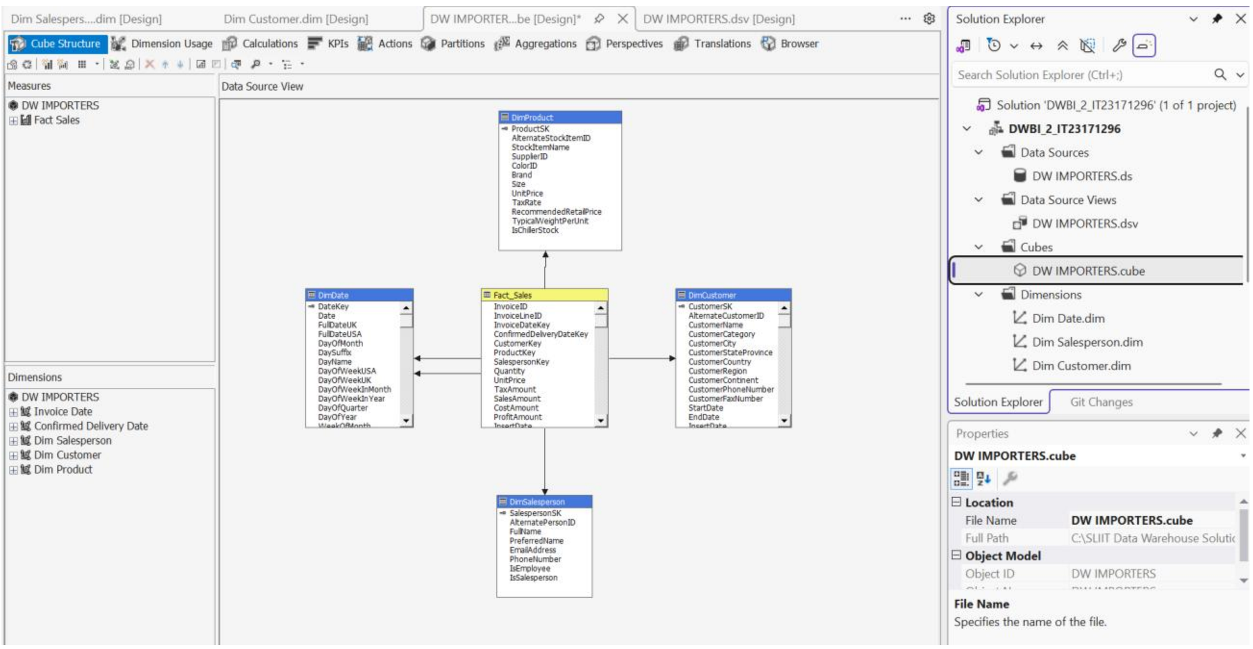
The DimProduct dimension was used to analyze product-related information. Attributes such as Product Name, Brand, Size, Color, and Product Category were included.

This dimension helps to evaluate sales performance by different product characteristics.



4.3 Create Cube

- The Cube Wizard made a cube with Fact_Sales as the measure group.and the numeric fields that have to do with sales as measures.
- We checked the use of dimensions and fixed any problems to make sure that all dimensions were connected to the fact measure group.
- The cube shows the main measures, such as Quantity, Sales Amount, Cost Amount, Profit Amount, Unit Price, Tax Amount and transaction processing time (in hours), if they are available.
- These measures let you add up numbers across any dimension and are the basis for Excel and Power BI analysis.



Hierarchies

• The cube design used hierarchies to make it easier for users to find their way around the data, which made it easier to analyze. These hierarchies make it easy to drill down -go from summary to detail and roll up - go from detail to summary.

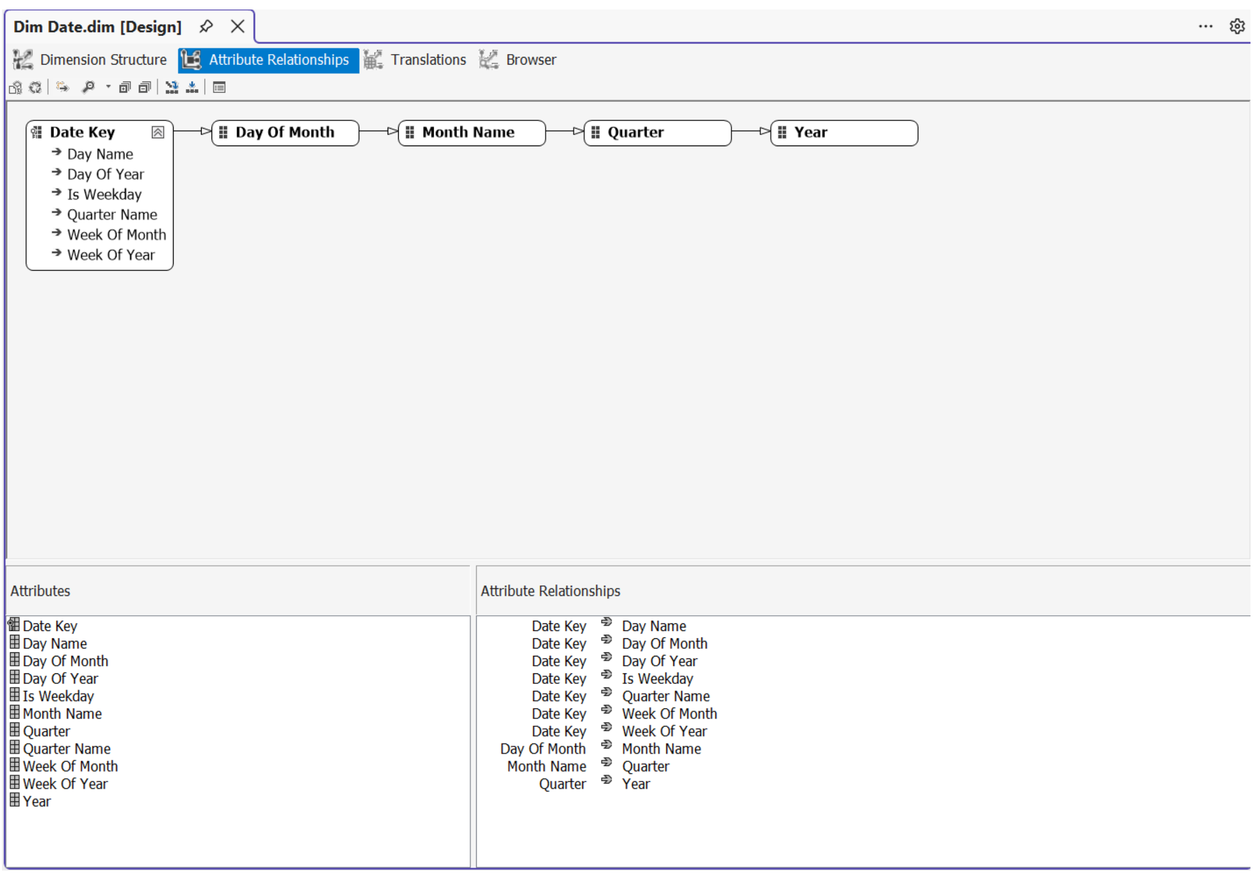
Date Hierarchy

• The Date hierarchy is the main hierarchy used in this project. It is set up like this:

Year → Quarter → Month Name → Day Of Month

• This hierarchy lets users look at data at different levels of time. For instance, users can start by looking at yearly sales trends, then go down to quarters, then to months, and finally to specific days.

• The Date Key is related to more qualities, such as Day Name, Day Of Year, Week Of Month, Week Of Year and Is Weekday. This makes it easier to filter and do thorough time-based analysis.



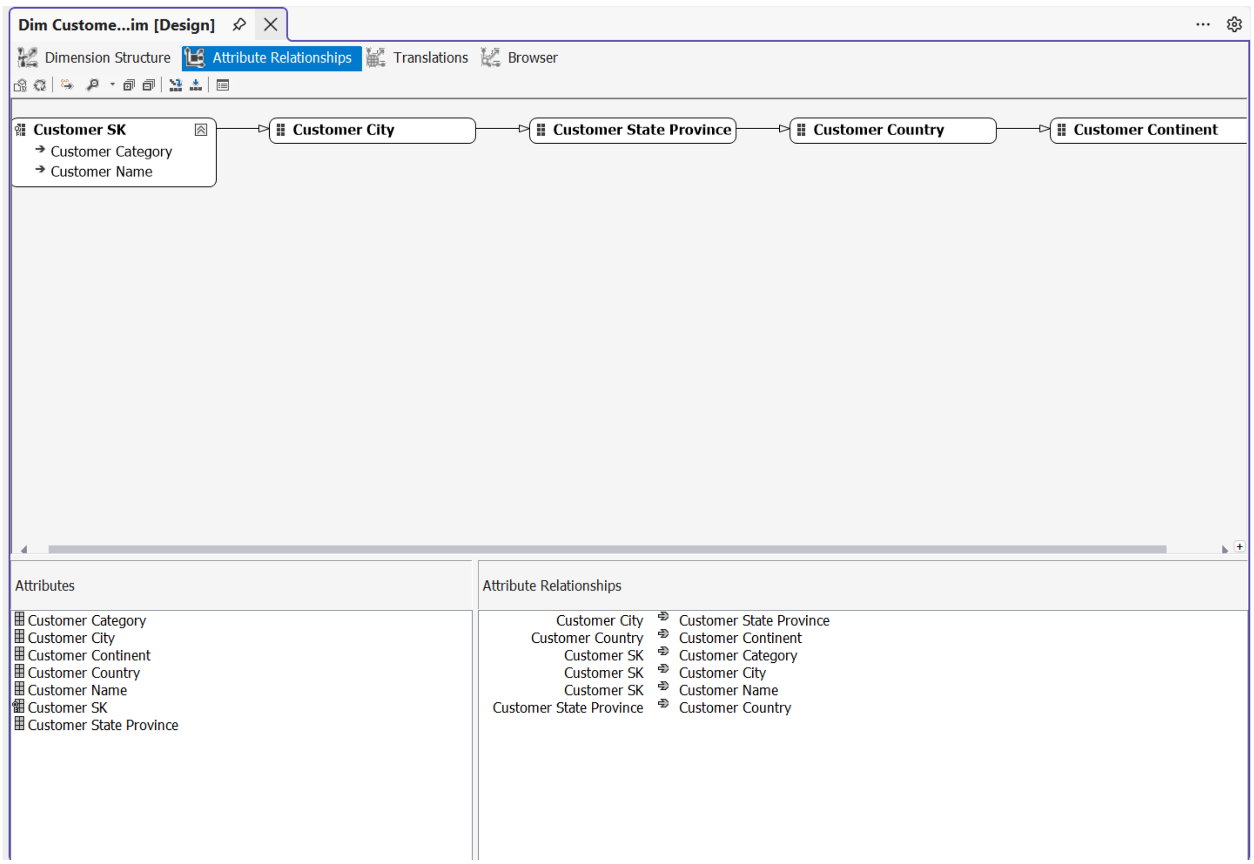
Customer Location Hierarchy

• A location hierarchy was added to the Customer dimension to make it easier to do geographical analysis. The hierarchy is set up like this:

Customer City -> Customer State Province → Customer Country → Customer Continent

• This hierarchy lets users look at sales data by location, starting with cities and moving up to larger areas like states, countries, and continents.

• The Customer Key is linked to attributes like Customer Name and Customer Category, which lets you look at data for each customer and for groups of customers.

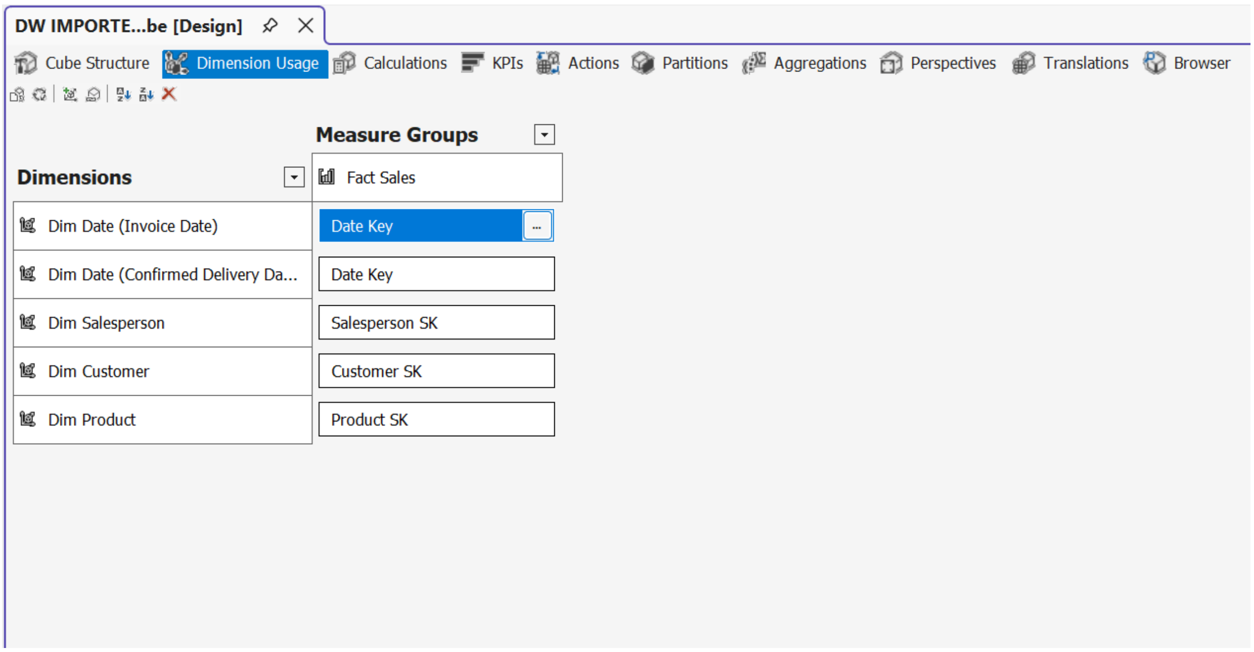


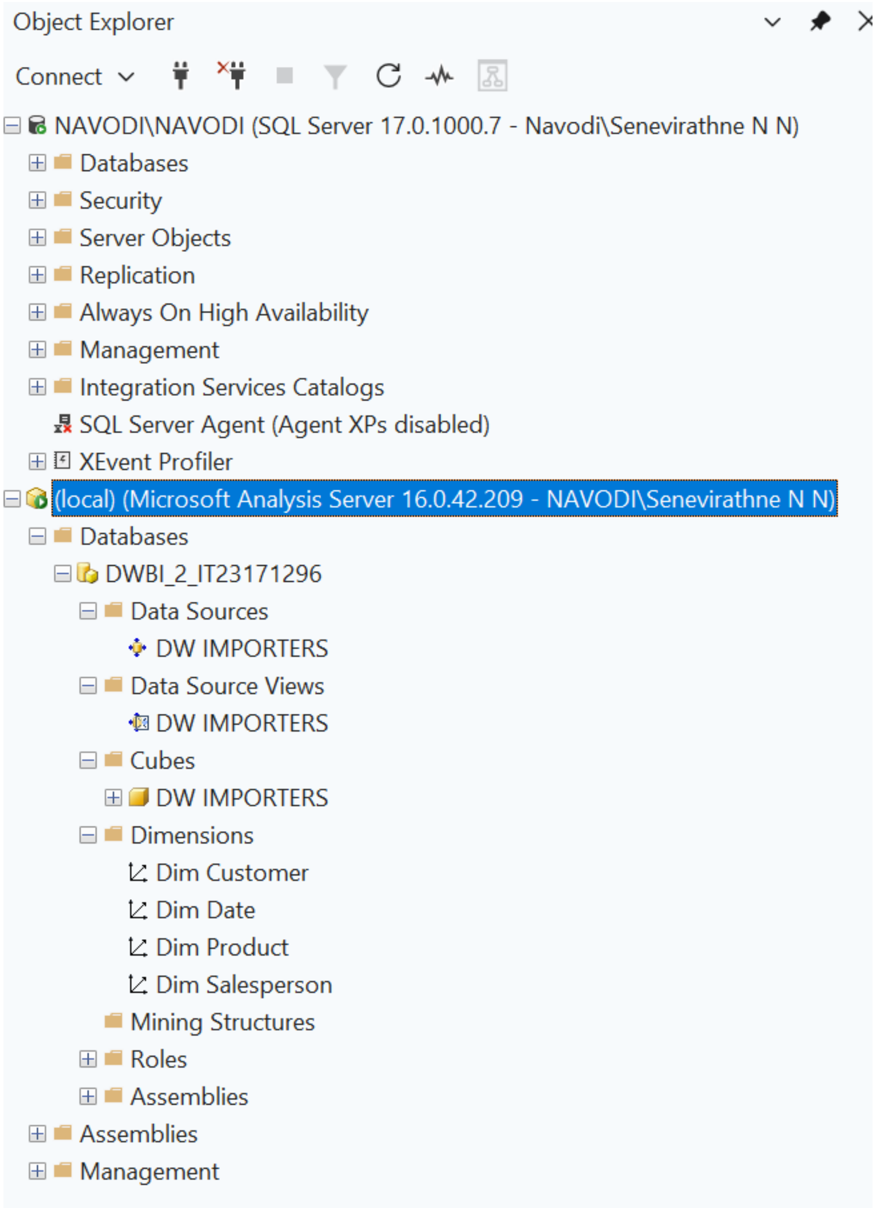
Overall Advantages

The cube's usability is enhanced by these hierarchies by,

- Putting characteristics into useful structures
- Facilitating easy navigation between various levels of detail
- Facilitating effective drill-down and roll-up procedures
- Improving analytical skills for location-based and time-based insights

All things considered, the design guarantees that users may effectively analyze sales data from high-level summaries to in-depth transactional views.





The cube processing was executed successfully, and the confirmation was also done by

expanding the DWBI_2_IT23271296 using SSMS.

We used SQL Server Analysis Services (SSAS) to make an OLAP cube. The cube combines fact and dimension tables and lets you analyze data in more than one way.
We set up measures like Sales Amount, Quantity, Profit Amount, and Cost Amount, and we made hierarchies for dimensions like Date, Customer and Product.

4.4 Measures and Aggregations

The cube includes several important measures.

- Sales Amount
- Cost Amount
- Profit Amount
- Quantity
- Tax Amount
- Unit Price

These measures are aggregated across dimensions to support business analysis.

Ex:

- Total Sales by country
- Profit by year
- Quantity by product category

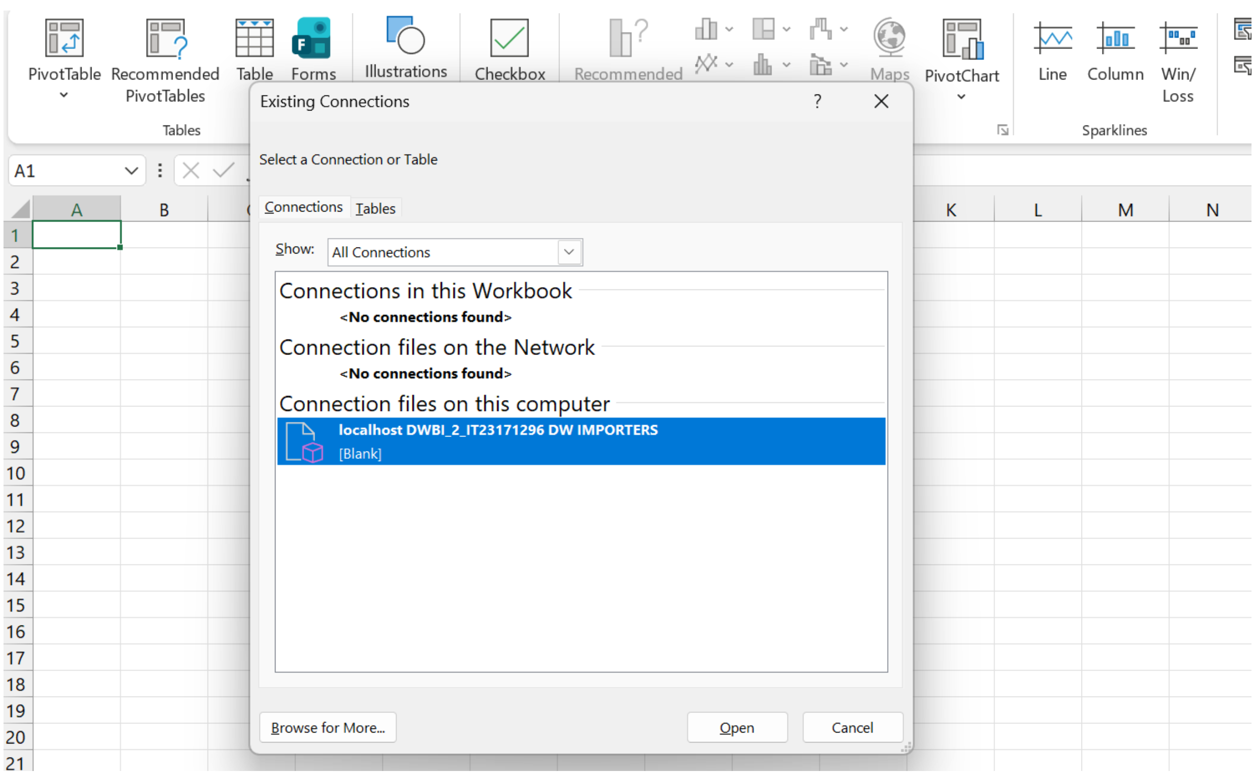
5. Excel OLAP Demonstration

We used PivotTables to analyze the processed cube after it was connected to Microsoft Excel. We used different worksheets to show the five OLAP operations that were needed: Roll-up, Drill-down, Slice, Dice, and Pivot

Linking Excel to the cube

Using the Data tab and an existing connection to Analysis Services, the Excel workbook was linked to the SSAS cube. The DW Importers cube was chosen as the source.

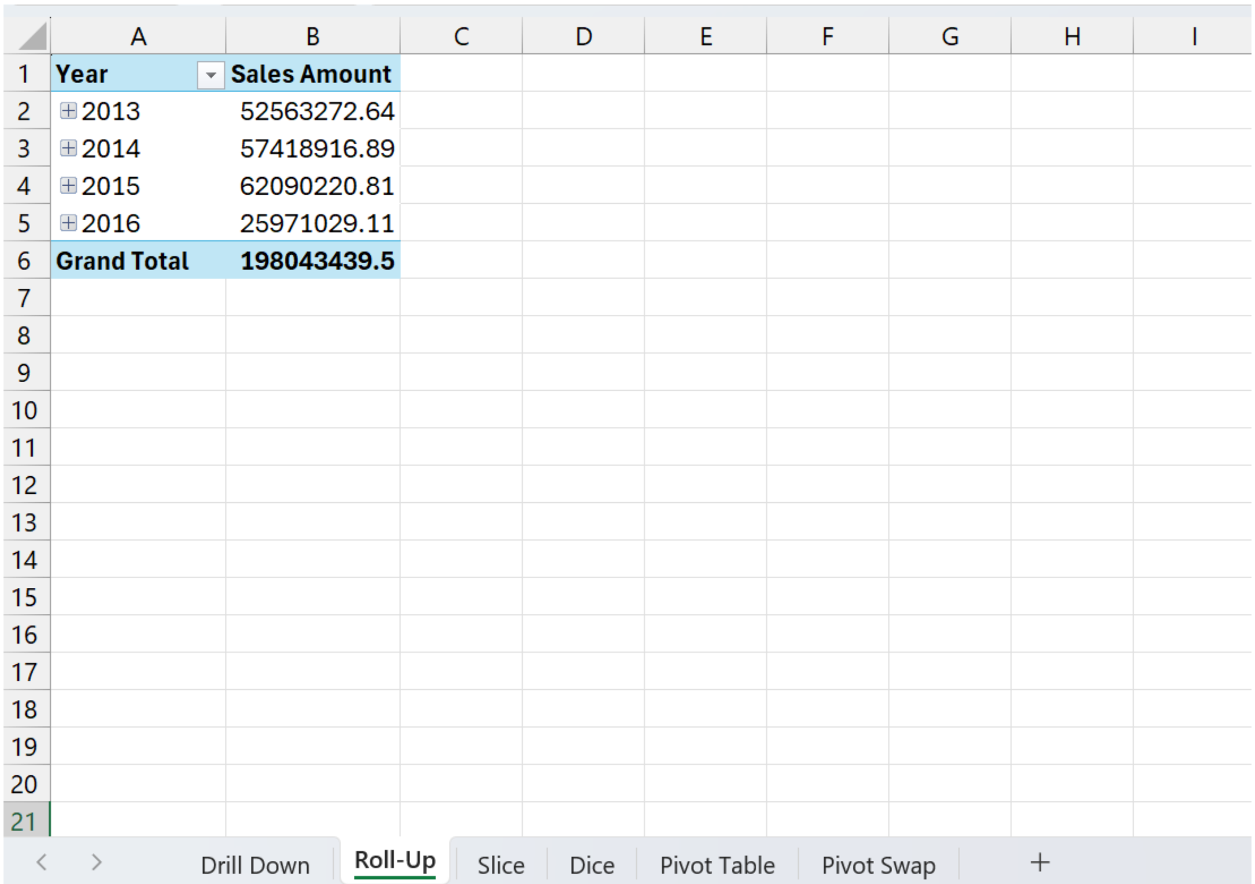
Once the connection was made, PivotTables were made in different worksheets so that each OLAP operation could be shown clearly.



5.1 Roll – up

Roll-up is an OLAP operation used to aggregate data to a higher level in a hierarchy.

Ex:
Viewing total sales by Year instead of Month.

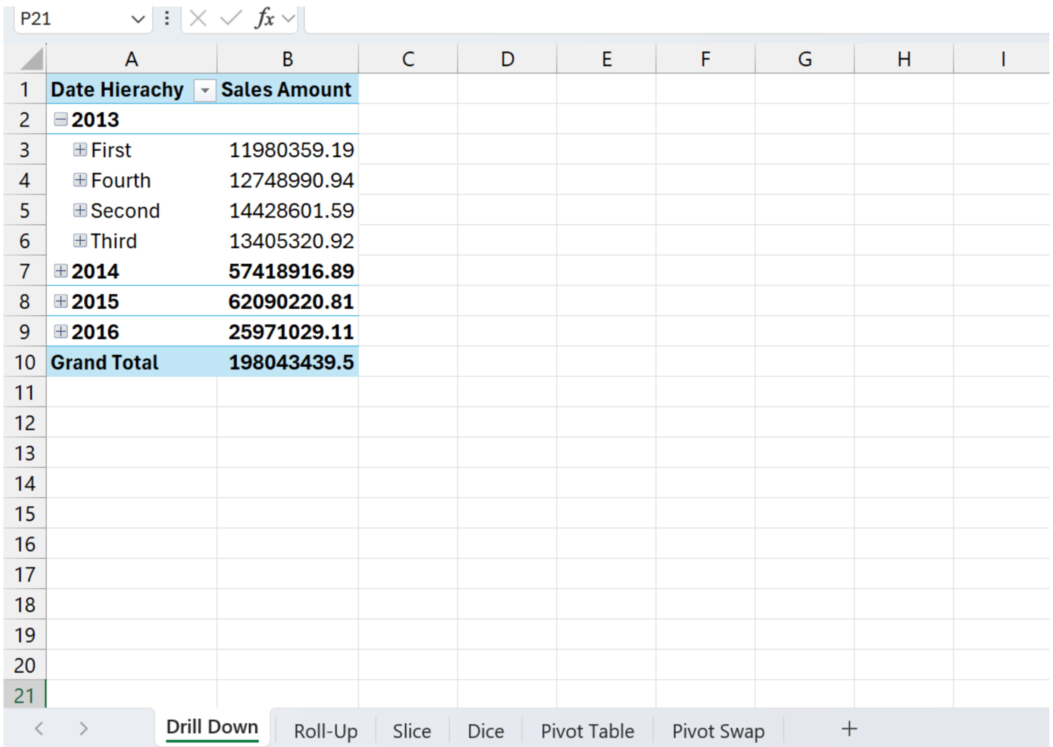


Year	Sales Amount
2013	52563272.64
2014	57418916.89
2015	62090220.81
2016	25971029.11
Grand Total	**198043439.5**

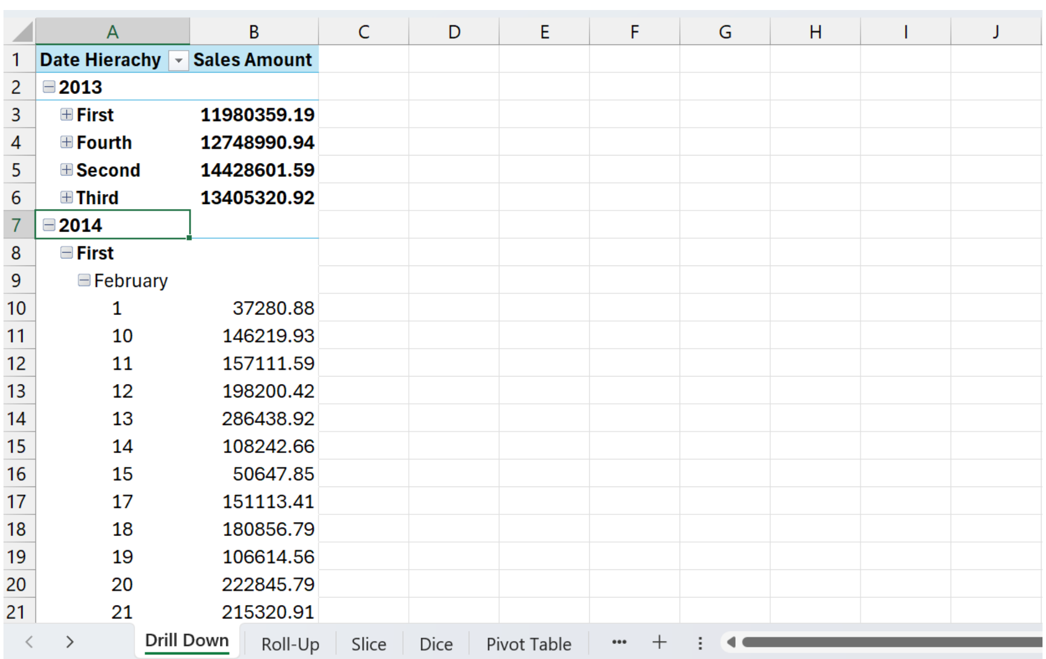
5.2 Drill – down

Roll-up is an OLAP operation used to aggregate data to a higher level in a hierarchy.

Ex:
Viewing total sales by Year instead of Month.



Date Hierachy	Sales Amount
2013	
First	11980359.19
Fourth	12748990.94
Second	14428601.59
Third	13405320.92
2014	**57418916.89**
2015	**62090220.81**
2016	**25971029.11**
Grand Total	**198043439.5**



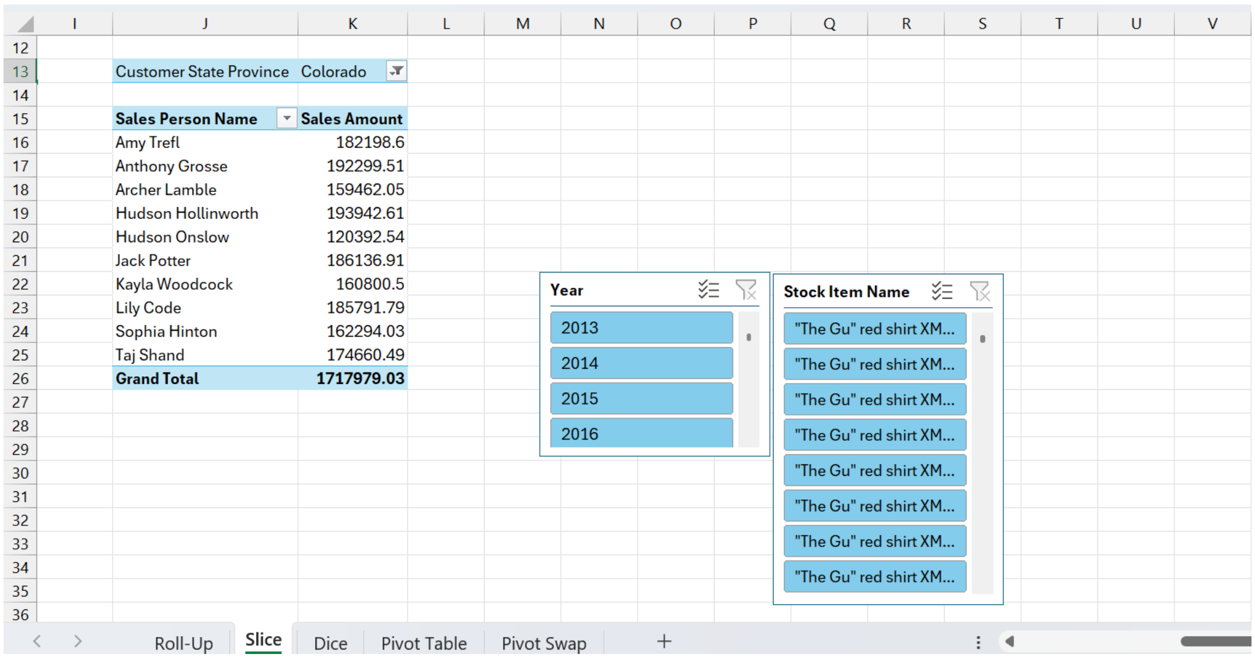
Date Hierachy	Sales Amount
2013	
First	**11980359.19**
Fourth	**12748990.94**
Second	**14428601.59**
Third	**13405320.92**
2014	
First	
February	
1	37280.88
10	146219.93
11	157111.59
12	198200.42
13	286438.92
14	108242.66
15	50647.85
17	151113.41
18	180856.79
19	106614.56
20	222845.79
21	215320.91

5.3 Slice

Dice operation filters data using multiple dimensions.
Ex:
Filtering by:

- Year = 2014
- Location = Colarado

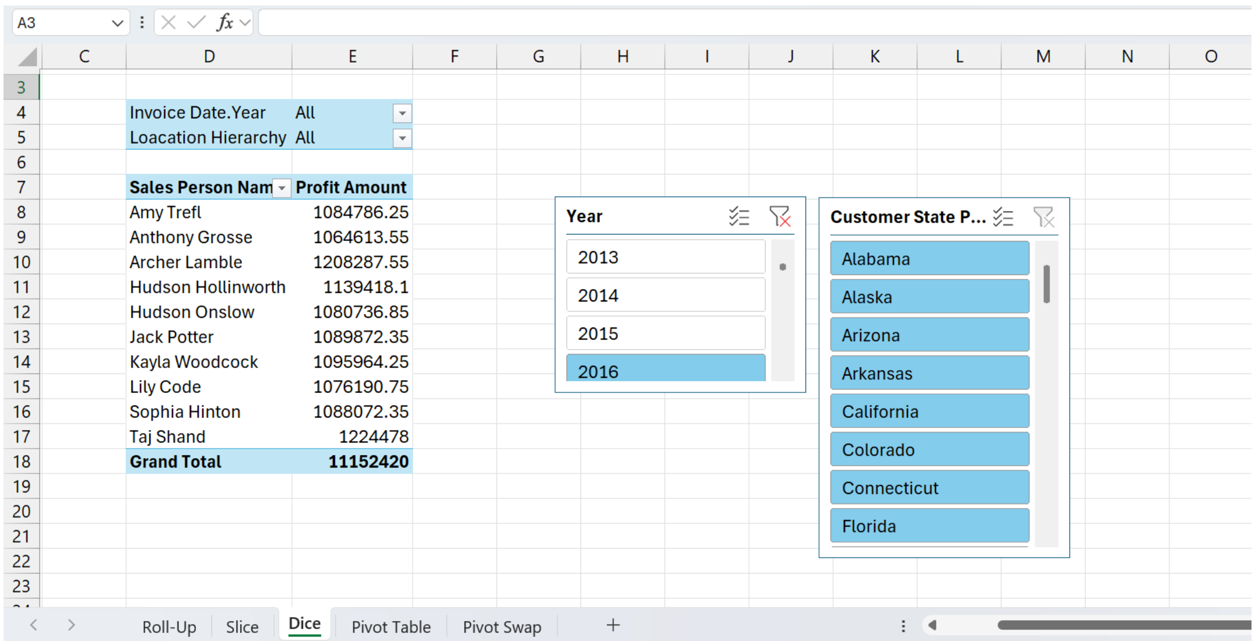
This provides a more detailed subset of the data.



5.4 Dice

Slice operation is used to filter data on a single dimension.

Ex:
Filtering by Customer State Province = Colorado shows sales values for that specific location.



Invoice Date.Year	All
Loacation Hierarchy	All

Sales Person Nam	Profit Amount
Amy Trefl	1084786.25
Anthony Grosse	1064613.55
Archer Lamble	1208287.55
Hudson Hollinworth	1139418.1
Hudson Onslow	1080736.85
Jack Potter	1089872.35
Kayla Woodcock	1095964.25
Lily Code	1076190.75
Sophia Hinton	1088072.35
Taj Shand	1224478
Grand Total	**11152420**

5.5 Pivot

Pivot is used to rearrange data by rotating rows and columns to view data from different perspectives.

Ex:
Switching Year from rows to columns to compare sales across years.

Sales Amount	Year				
State Province name	**2013**	**2014**	**2015**	**2016**	**Grand Total**
Alabama	34665485.37	36347336.8	38106611.2	15818990.5	124938424
Alaska	420138.77	431099.77	553081.06	211738.76	1616058.36
Arizona	255986.09	249857.07	283916.75	126821.11	916581.02
Arkansas	408186.67	487001.62	611606.84	319275.51	1826070.64
California	1063861.99	1060445.96	1215705.55	490141.96	3830155.46
Colorado	340563.17	535943.12	625222.85	216249.89	1717979.03
Connecticut			91046.96	76489.51	167536.47
Florida	655099.17	658669.95	833119.95	423155.96	2570045.03
Georgia	247470.89	264687.14	533897.75	224744.65	1270800.43
Idaho	213226.49	293248.24	341690.56	115349.22	963514.51
Illinois	560810.21	713990.69	694682.17	416928.85	2386411.92
Indiana	242407.91	411745.25	368790.83	176357.93	1199301.92
Iowa	402142.63	517790.26	487246.94	260891.5	1668071.33
Kansas	438290.4	528210.46	503468.75	272366.23	1742335.84
Kentucky	471105.47	462591.58	485657.87	131461.55	1550816.47
Louisiana	483583.22	444360.27	594201.83	205140.66	1727285.98
Maine	34598.28	120416.06	175846.57	93001.49	423862.4
Maryland	130693.32	215819.35	205545.14	69855.88	621913.69
Massachusetts[E]	585175.73	682175.67	740823.28	379082.75	2387257.43
Michigan	212461.03	246436.7	305896.99	137841.2	902635.92
Minnesota	639883.83	594509.12	856096.78	298341.85	2388831.58
Mississippi	131138.03	167123.94	223498.68	89185.84	610946.49
Missouri	409081.2	551960.58	565252.4	237587.61	1763881.79
Montana	136533.53	173917.06	321886.12	153972.04	786308.75
Nebraska	379489.77	415347.78	431743.64	144272.57	1370853.76
Nevada	48297.01	105935.24	91839.54	21090.2	267161.99
New Hampshire		40444.58	90837.53	84087.68	215369.79
New Jersey	559259.96	665983.48	789709.91	248631.5	2263584.85
New Mexico	368753.23	528861.88	379744.12	144681.24	1422040.47
New York	826165.45	979372.12	1071828.44	545719.52	3423085.53
North Carolina	679139.72	925044.61	965369.11	429128.03	2998681.47
North Dakota	141691.25	191661.61	201992.24	65057.31	600402.41
Ohio	640285.77	612274.6	738621.19	287700.86	2278882.42
Oklahoma	498224.7	701971.79	988590.06	331529.55	2520316.1
Oregon	190418.06	248551.2	358023.65	147496.03	944488.94
Pennsylvania	880836.21	989919.77	1055994.01	458236.47	3384986.46
Puerto Rico (US Territory)	95327.65	86072.85	168687.02	61694.1	411781.62
South Carolina	373542.15	396535.92	570256.69	245915.69	1586250.45
South Dakota	384448.79	428198.27	368387.72	154141.39	1335176.17
Tennessee	111198.92	138203.4	193643.75	48679.35	491725.42
Texas	1443154.16	1544842.36	1608412.24	585743.44	5182152.2
Utah	63204.93	92964.1	170521.5	88442.07	415132.6
Vermont	106611.42	223826	130521.67	56286.19	517245.28
Virginia	298059.27	400376.1	350222.58	229931.53	1278589.48
Washington	265568.54	373275.41	372021.95	113145.68	1124011.58
West Virginia	572064.12	620881.99	668688.2	269585.92	2131220.23
Wisconsin	203899.29	293647.92	317593.69	157461.72	972602.62
Wyoming	285708.87	255387.25	282176.54	107398.64	930671.3
Grand Total	**52563272.64**	**57418916.9**	**62090220.8**	**25971029.1**	**198043439**

Drill Down | Roll-Up | Slice | Dice | Pivot Table | Pivot Swap

Sales Amount	State Province													
Year	Alabama	Alaska	Arizona	Arkansas	California	Colorado	Connecticut	Florida	Georgia	Idaho	Illinois	Indiana	Iowa	Kansas
2013	34665485.37	420138.77	255986.09	408186.67	1063861.99	340563.17		655099.17	247470.89	213226.49	560810.21	242407.91	402142.63	438290.4
2014	36347336.8	431099.77	249857.07	487001.62	1060445.96	535943.12		658669.95	264687.14	293248.24	713990.69	411745.25	517790.26	528210.46
2015	38106611.2	553081.06	283916.75	611606.84	1215705.55	625222.85	91046.96	833119.95	533897.75	341690.56	694682.17	368790.83	487246.94	503468.75
2016	15818990.48	211738.76	126821.11	319275.51	490141.96	216249.89	76489.51	423155.96	224744.65	115349.22	416928.85	176357.93	260891.5	272366.23
Grand Total	**124938423.9**	**1616058.36**	**916581.02**	**1826070.64**	**3830155.46**	**1717979.03**	**167536.47**	**2570045.03**	**1270800.43**	**963514.51**	**2386411.92**	**1199301.92**	**1668071.33**	**1742335.84**

Drill Down | Roll-Up | Slice | Dice | Pivot Table | Pivot Swap

We used pivot tables in Microsoft Excel to look at the SSAS cube data. Users could drag and drop fields for analysis because the cube was linked to Excel.
The pivot table shows how sales in the United States have changed over time, showing how different measures can be combined and studied.

6. Power BI Report

Power BI was connected to the SSAS cube in order to produce dynamic and interactive dashboards. Real-time data analysis was made possible without the need to import data thanks to the establishment of a live link.

Visual insights into sales performance across various dimensions, including time, location, customer and product are provided by the reports.

• Graphs of bars
• Visuals of the matrix
• Slicers and filters

These visuals let people interact with data in real time.

6.1 DAX Measures

Since the Power BI reports were connected using a live connection to the SSAS cube, most of the required measures were already defined during cube implementation.

Measures such as

- Sales Amount
- Profit Amount
- Cost Amount
- Quantity

were reused directly from the cube.

Therefore, additional DAX measures were not required in Power BI. This approach ensures consistency and avoids duplication of calculations across tools.

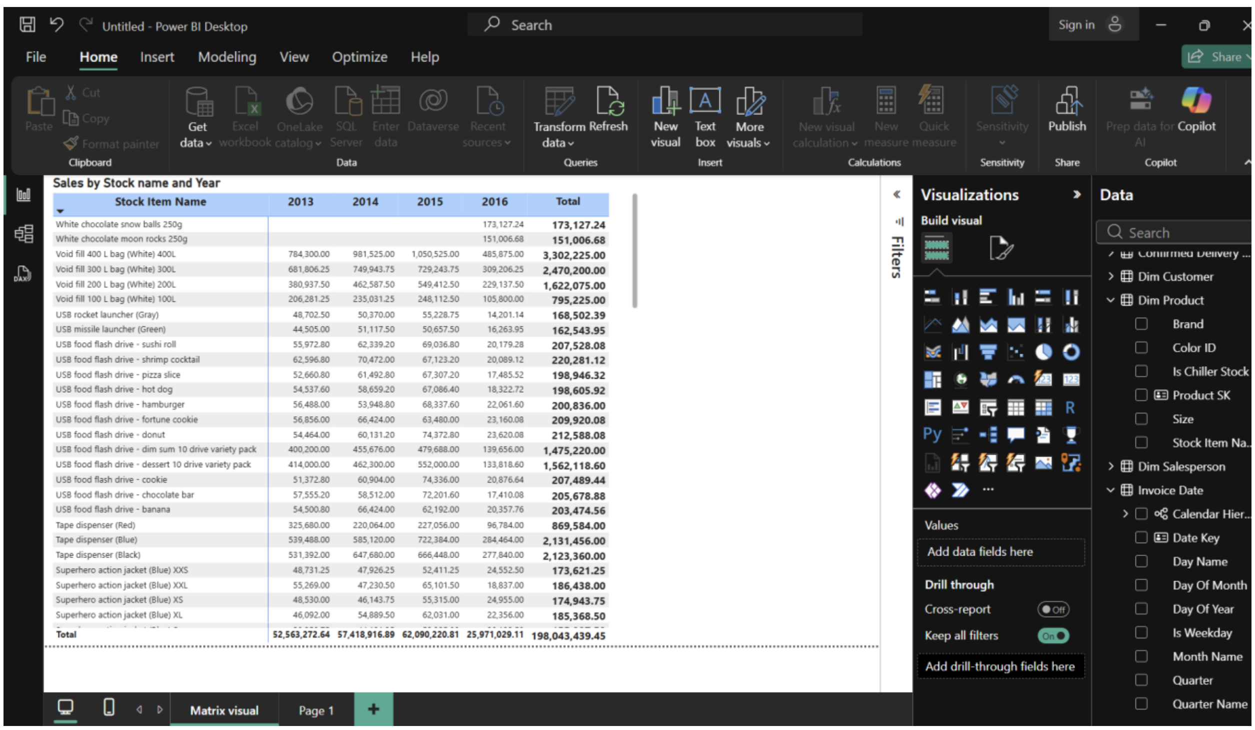
6.2 Power BI reports

Report 1 - Matrix Visual

To show structured tabular data with row and column groupings, a matrix visual was made.

Rows: Product Category/Customer Country
Columns: Year
Values: Total Sales

This visual allows users to compare sales performance across multiple dimensions simultaneously



Report 2 - Multiple Slicers and Visual Insights

The implementation of multiple slicers allowed for dynamic data filtering.

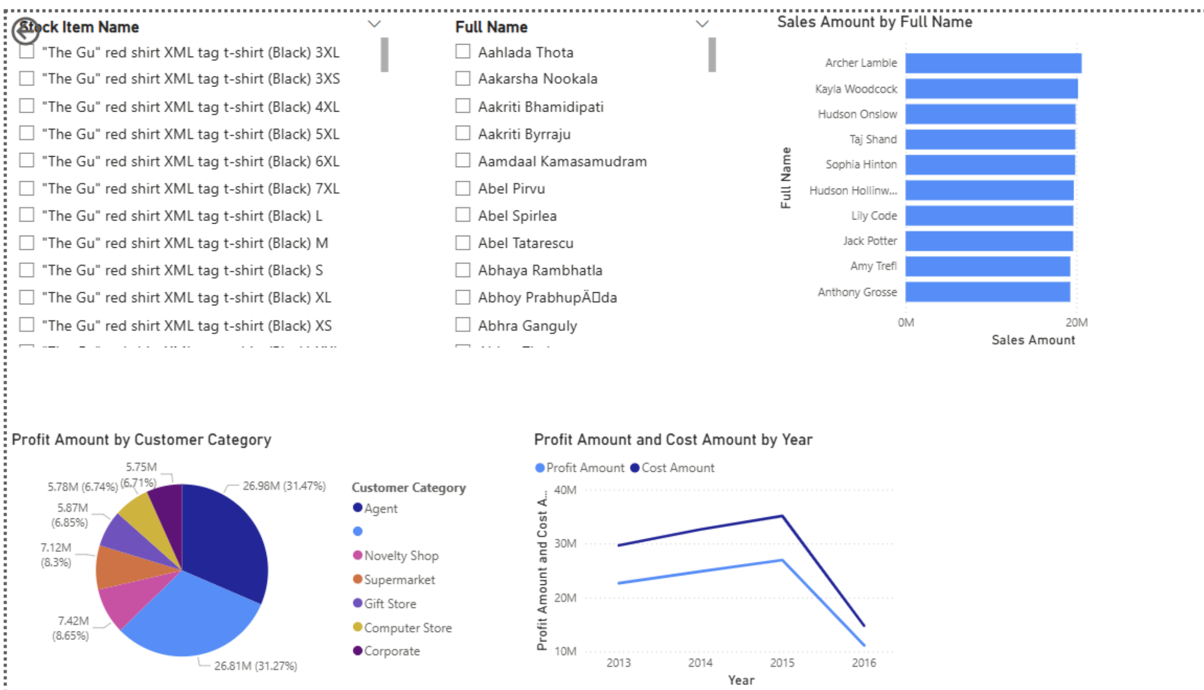
- Year
- Customer Nation
- Product Type

Are the slices that used.

Because these slicers are linked (cascading filters), choosing one filter dynamically modifies the options available in other slicers.

Other images utilized:

- Line chart → Sales trend over time
- Bar chart → Sales by Country
- Pie graphic → Distribution of Sales

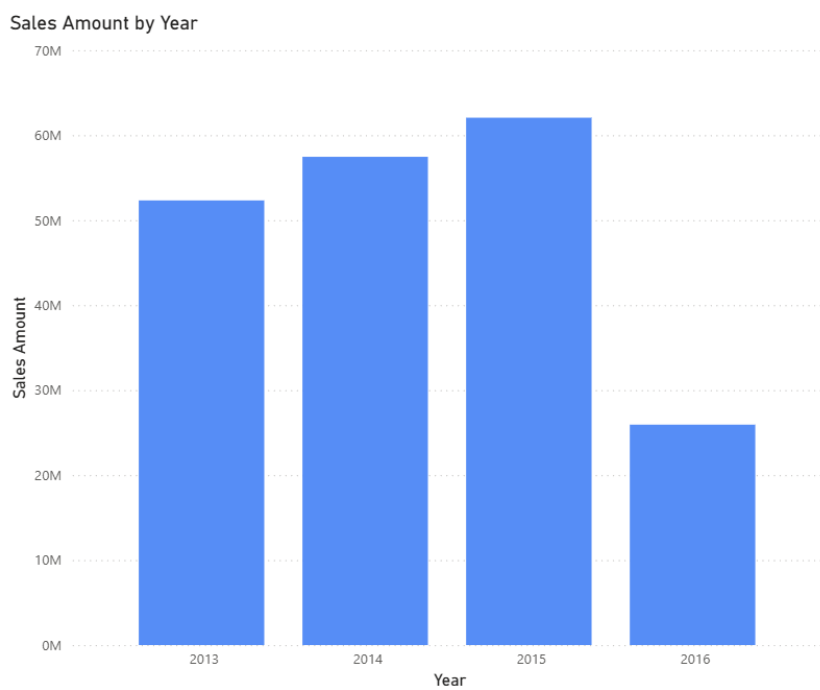


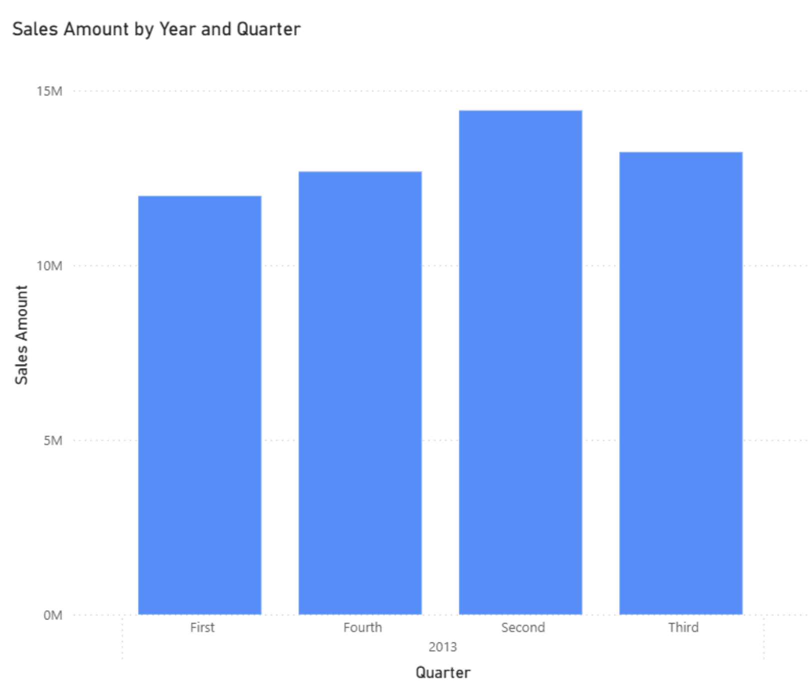
Report 3 - Drill-Down Report

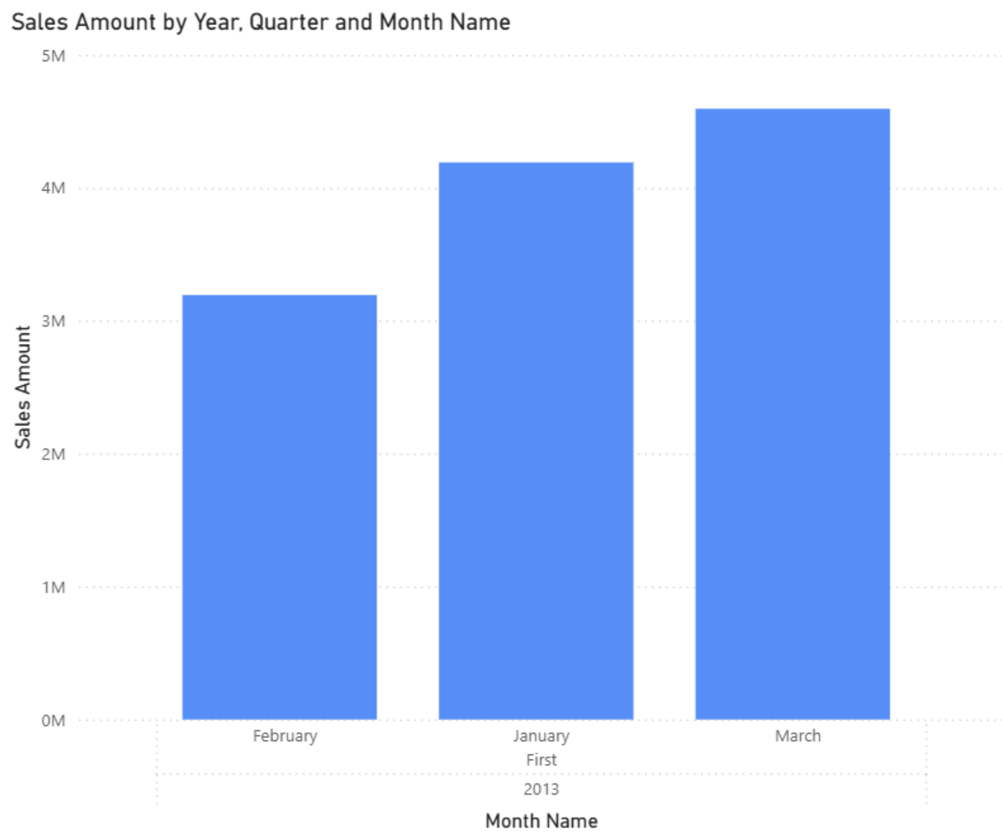
Hierarchies were used to develop drill-down functionality.

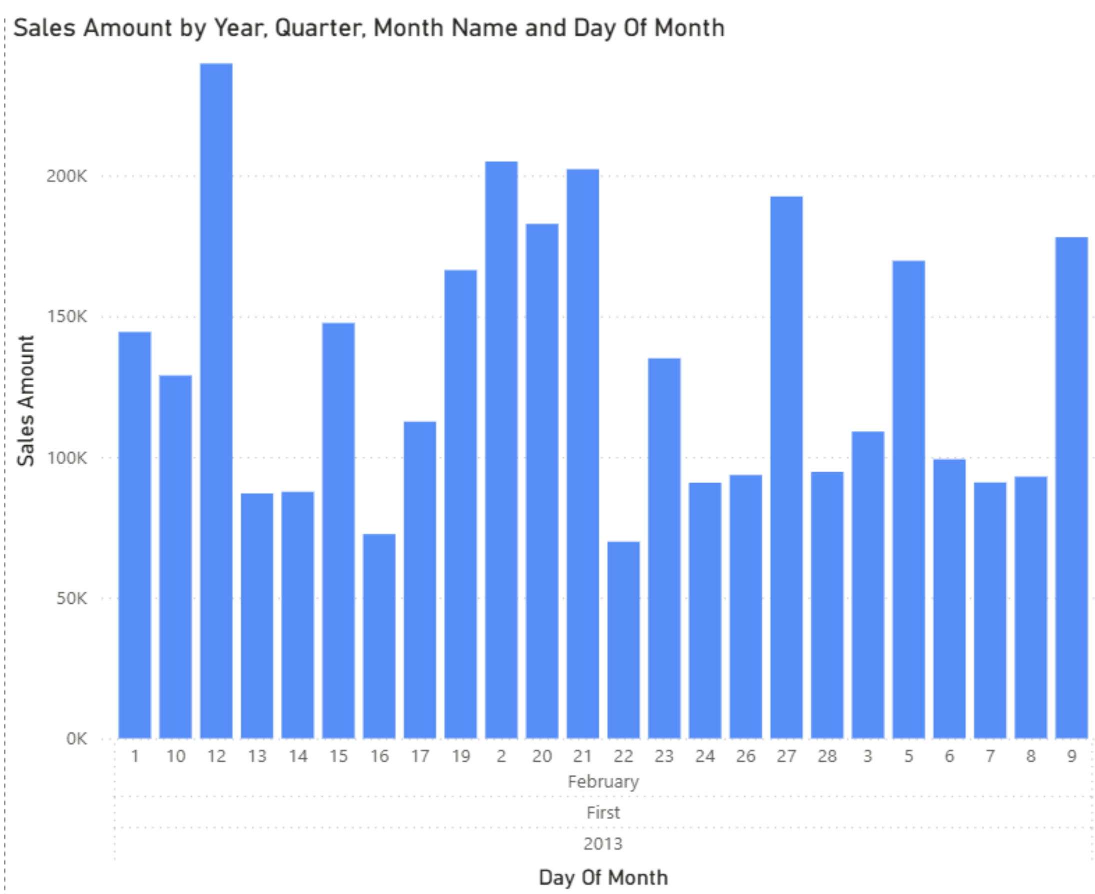
Ex:
Month ← Day → Quarter → Year

Clicking on images allows users to examine data at various levels of detail.
This enhances analytical capacity by enabling in-depth trend analysis.






Sales Amount by Year, Quarter and Month Name
5M
4M
3M
2M
1M
0M
Sales Amount
February
January
March
First
2013
Month Name


Sales Amount by Year, Quarter, Month Name and Day Of Month
200K
150K
100K
50K
0K
Sales Amount
1
10
12
13
14
15
16
17
19
2
20
21
22
23
24
26
27
28
3
5
6
7
8
9
February
First
2013
Day Of Month

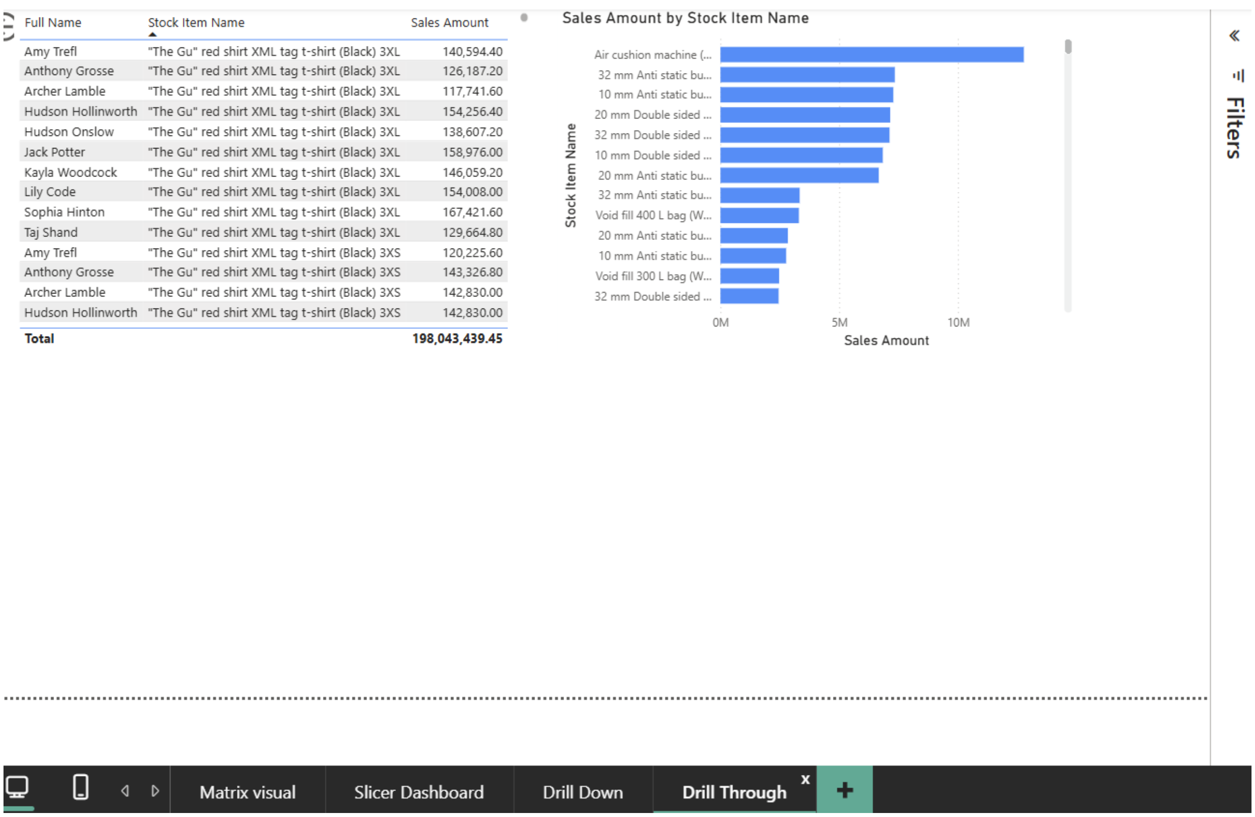
Report 4 - Drill-Through Report

Drill-through enables access to a comprehensive report page.

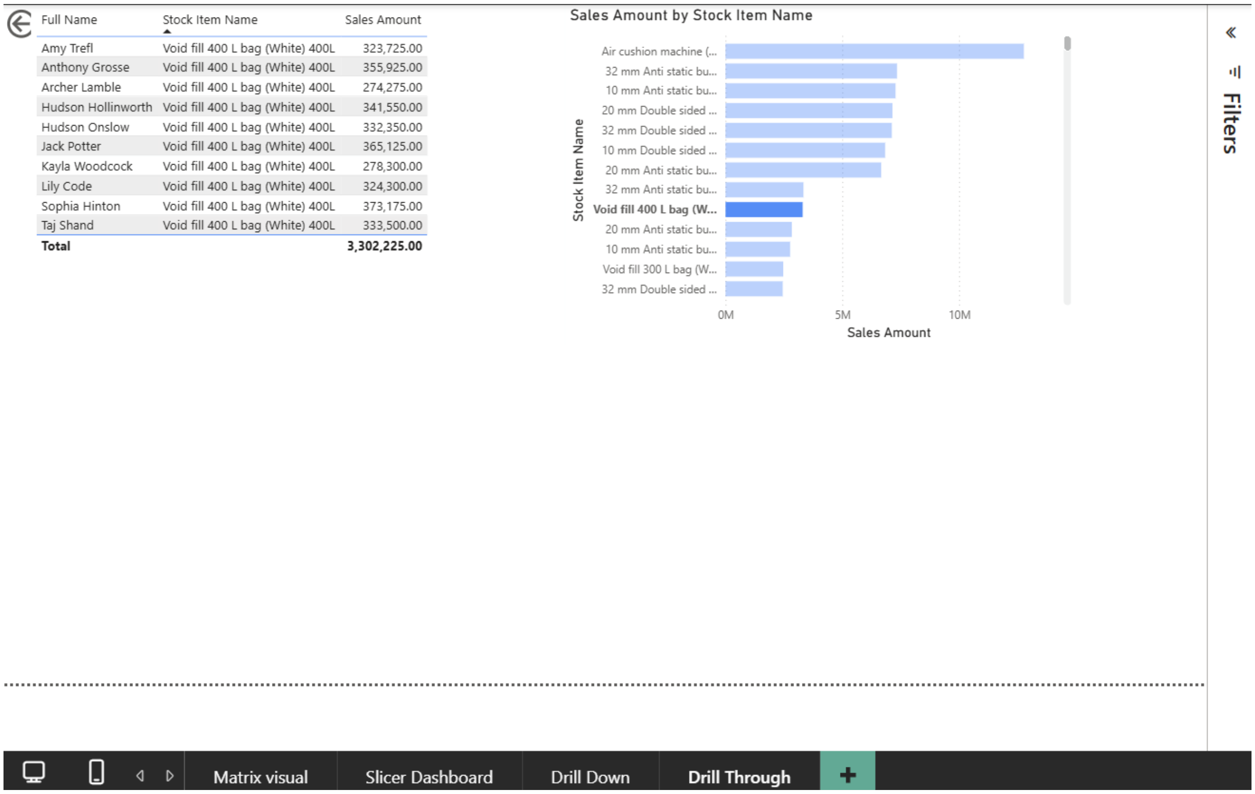
Steps,

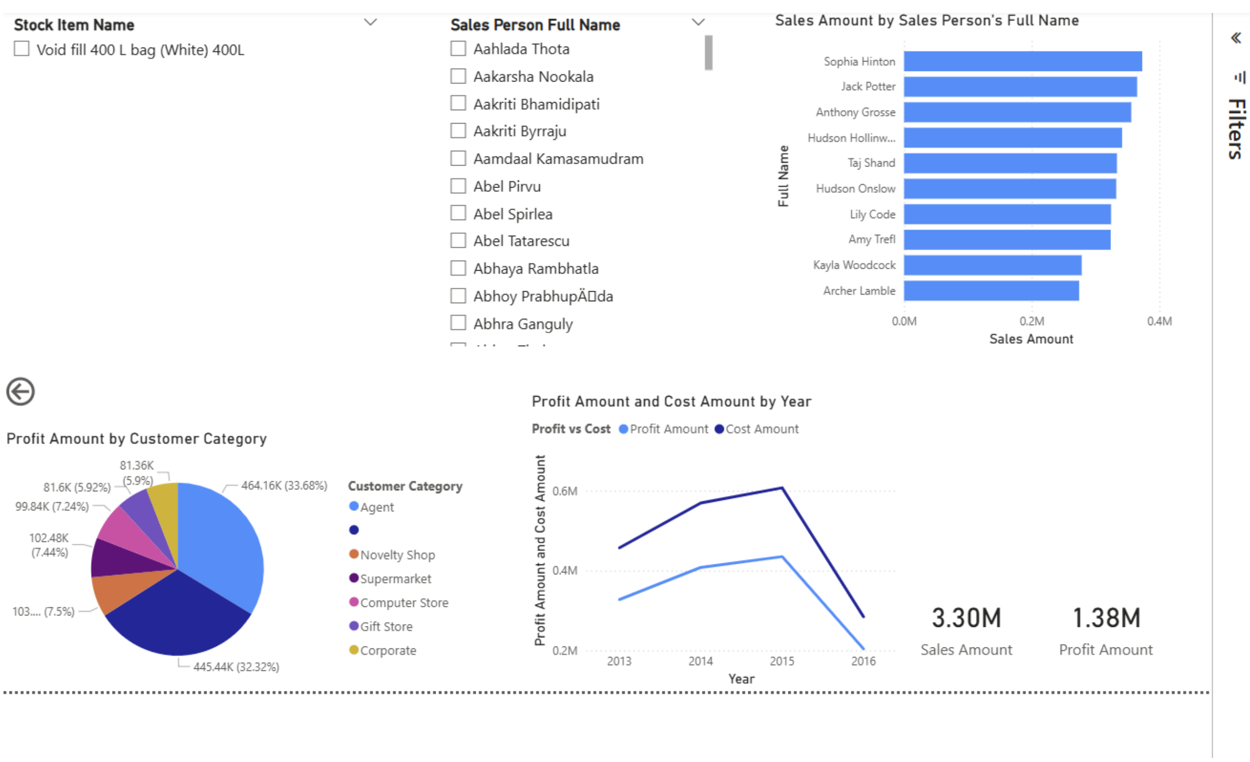
- developed a different page called the Detail Page.
- Drill-through filter addition (e.g., Customer Country)
- Turned on right-click navigation

A chart's detailed information for a chosen value can be viewed by right-clicking on it.



Full Name	Stock Item Name	Sales Amount
Amy Trefl	"The Gu" red shirt XML tag t-shirt (Black) 3XL	140,594.40
Anthony Grosse	"The Gu" red shirt XML tag t-shirt (Black) 3XL	126,187.20
Archer Lamble	"The Gu" red shirt XML tag t-shirt (Black) 3XL	117,741.60
Hudson Hollinworth	"The Gu" red shirt XML tag t-shirt (Black) 3XL	154,256.40
Hudson Onslow	"The Gu" red shirt XML tag t-shirt (Black) 3XL	138,607.20
Jack Potter	"The Gu" red shirt XML tag t-shirt (Black) 3XL	158,976.00
Kayla Woodcock	"The Gu" red shirt XML tag t-shirt (Black) 3XL	146,059.20
Lily Code	"The Gu" red shirt XML tag t-shirt (Black) 3XL	154,008.00
Sophia Hinton	"The Gu" red shirt XML tag t-shirt (Black) 3XL	167,421.60
Taj Shand	"The Gu" red shirt XML tag t-shirt (Black) 3XL	129,664.80
Amy Trefl	"The Gu" red shirt XML tag t-shirt (Black) 3XS	120,225.60
Anthony Grosse	"The Gu" red shirt XML tag t-shirt (Black) 3XS	143,326.80
Archer Lamble	"The Gu" red shirt XML tag t-shirt (Black) 3XS	142,830.00
Hudson Hollinworth	"The Gu" red shirt XML tag t-shirt (Black) 3XS	142,830.00
Total		**198,043,439.45**


Full Name
Stock Item Name
Sales Amount
Amy Trefl Void fill 400 L bag (White) 400L 323,725.00
Anthony Grosse Void fill 400 L bag (White) 400L 355,925.00
Archer Lamble Void fill 400 L bag (White) 400L 274,275.00
Hudson Hollinworth Void fill 400 L bag (White) 400L 341,550.00
Hudson Onslow Void fill 400 L bag (White) 400L 332,350.00
Jack Potter Void fill 400 L bag (White) 400L 365,125.00
Kayla Woodcock Void fill 400 L bag (White) 400L 278,300.00
Lily Code Void fill 400 L bag (White) 400L 324,300.00
Sophia Hinton Void fill 400 L bag (White) 400L 373,175.00
Taj Shand Void fill 400 L bag (White) 400L 333,500.00
Total 3,302,225.00
Sales Amount by Stock Item Name
Air cushion machine (...
32 mm Anti static bu...
10 mm Anti static bu...
20 mm Double sided ...
32 mm Double sided ...
10 mm Double sided ...
20 mm Anti static bu...
32 mm Anti static bu...
Void fill 400 L bag (W...
20 mm Anti static bu...
10 mm Anti static bu...
Void fill 300 L bag (W...
32 mm Double sided ...
Stock Item Name
0M
5M
10M
Sales Amount
Filters
Matrix visual
Slicer Dashboard
Drill Down
Drill Through


Stock Item Name
Void fill 400 L bag (White) 400L
Sales Person Full Name
Aahlada Thota
Aakarsha Nookala
Aakriti Bhamidipati
Aakriti Byrraju
Aamdaal Kamasamudram
Abel Pirvu
Abel Spirlea
Abel Tatarescu
Abhaya Rambhatla
Abhoy PrabhupÄda
Abhra Ganguly
Sales Amount by Sales Person's Full Name
Sophia Hinton
Jack Potter
Anthony Grosse
Hudson Hollinw...
Taj Shand
Hudson Onslow
Lily Code
Amy Trefl
Kayla Woodcock
Archer Lamble
Full Name
0.0M
0.2M
0.4M
Sales Amount
Filters
Profit Amount by Customer Category
81.36K (5.9%)
81.6K (5.92%)
99.84K (7.24%)
102.48K (7.44%)
103.... (7.5%)
464.16K (33.68%)
445.44K (32.32%)
Customer Category
Agent
Novelty Shop
Supermarket
Computer Store
Gift Store
Corporate
Profit Amount and Cost Amount by Year
Profit vs Cost
Profit Amount
Cost Amount
Profit Amount and Cost Amount
0.6M
0.4M
0.2M
2013
2014
2015
2016
Year
3.30M
Sales Amount
1.38M
Profit Amount

6.3 KPIs and Business Insights

The investigation revealed the following Key Performance Indicators (KPIs):

- Total Revenue from Sales
- Total Sales Revenue
- Total Profit
- Sales Growth by Year
- Top Performing Countries
- Best Selling Products

Best Selling Products Business Insights:

- A substantial portion of total sales revenue comes from the United States.
- From 2013 to 2015 , sales trended upward and in 2016 tended to downward
- Some areas function poorly and need to be strategically improved.
- Different product categories have different profit margins, which suggests chances for cost minimization.

Conclusion

This assignment effectively showcases the end-to-end Business Intelligence solution using a data warehouse, SSAS cube and Power BI.

Star schema was used to enhance data structuredness and query speeds, and the SSAS cube for multidimensional analytics. Interactive and easy visualizations using power BI

Overall, the system supports:

- Efficient data analysis
- OLAP operations
- Interactive reporting and decision-making